

Practical Task Sheet 3

ETL Process in the Data Warehouse

The fictitious beverage market chain *DrinkMix* will continue to serve as an example scenario for the practical task.

Implement the necessary loading and transformation steps for both tasks in SQL or PL/SQL! (Do NOT do it manually by hand!) Implementations in other programming languages should have at least a connection to the DBMS. Perform the given tasks in your DBS account and submit your code to Moodle before the deadline!

Task Description

In this last task, the data from the analysed branch databases shall be loaded into the data warehouse. Here you will not just use simple data integration and normalization, but also duplicate detection.

Below is an overview of the things to do in this task:

1. Read customers, normalize their attributes and clean them.
2. Detect duplicate customer tuples, i. e. when a customer is registered in multiple branches and would thus be multiple times in your database, if you simply copied the data.
3. Read products, normalize and clean their attributes.
4. Calculate the turnover (sales volume)
5. Calculate the amount of sales in liters for the turnover calculation
6. Determine the age groups of customers

Once found mappings of customers to branches should be saved separately. This way the mapping stays consistent and can be used to simplify repeated loading and/or duplicate detection procedures.

Further advice:

- Assume that there was never an address change of a customer, i. e. the same customer always has the same address, no matter the branch or time.
- Every customer is only once registered per branch, i. e. inside a branch you do not have to find duplicates.